

An Independent, Unlensed Gas-Phase Metallicity Deficit at $z \simeq 9\text{--}10$, Robust to the Local Mass–Metallicity Anchor

NEBULAMIND LAB (AUTONOMOUS RUN)¹

¹*Descriptive, machine-generated draft — not a validated measurement, and not a detection claim.*

ABSTRACT

Whether the earliest galaxies are already chemically enriched or remain genuinely metal-poor relative to the local mass–metallicity relation (MZR) is unsettled. JWST/NIRSpec studies have variously reported a declining MZR normalization toward $z \approx 8$ (Langeroodi et al. 2023; Sarkar et al. 2025) and rapid early enrichment that keeps massive galaxies near local abundances by $z \approx 5$ (Faisst et al. 2026), and the earliest $z > 7$ anchors that would break the tie are dominated by gravitationally lensed galaxies — where differential magnification distorts the inferred stellar masses, and where the (fundamental) metallicity relation has historically been probed (cf. Belli et al. 2013). We adjudicate this by measuring the offset of $z > 7$ star-forming galaxies from the local MZR on a single, Te-consistent abundance scale, separating lensed from genuinely unlensed field samples. Restricting the Nakajima et al. (2023) compilation to direct electron-temperature (T_e) abundances yields a deficit of -0.47 ± 0.10 dex, but two of its anchors (ERO, GLASS) lie behind lensing clusters. Using instead the strictly *unlensed* field sample of Pollock et al. (2026) (CAPERS/JADES, $N = 5$ direct- T_e , $z = 9.3\text{--}9.9$, $\log M_\star = 8.2\text{--}8.6$), we recover -0.69 ± 0.03 dex, robust to leave-one-out (spread 0.04 dex). The deficit is also robust to the choice of *local* anchor: replacing the Curti et al. (2020) relation — extrapolated below its SDSS calibration mass range — with the direct- T_e stacked MZR of Andrews & Martini (2013), which is *measured* at these masses, changes the deficit by only 0.04 dex (to -0.65 dex). An independent, much larger stacked- T_e sample — the ~ 1500 -galaxy JADES analysis of Isobe et al. (2026) — gives a consistent normalization deficit of -0.5 to -0.6 dex at $\log M_\star = 8$ ($12 + \log \text{O}/\text{H} = 7.62$ at $\log M_\star = 8$ over $z = 4\text{--}10$) via a different method; the $z \simeq 9\text{--}10$ -specific value rests on the individual detections. Within the unlensed sample the offset shows no significant trend with stellar mass or redshift — a pure normalization deficit at unchanged slope. The dominant remaining uncertainty is the absolute T_e abundance scale ($\sim 0.1\text{--}0.2$ dex). On an independent, unlensed, single-scale footing the data therefore land on the metal-poor side of this debate: a robust deficit relative to the local MZR, present across two high- z samples and two local anchors — and explicitly *not* a formal statistical detection, nor a validated measurement. What it settles is the sign and approximate size of the $z \simeq 9\text{--}10$ offset on lensing-free, single-scale data; what it cannot yet settle is the precise value, bounded by the $\sim 0.1\text{--}0.2$ dex T_e -scale floor and by the small unlensed individual-detection sample ($N = 5$, or 6 including GN-z11 at $z = 10.6$; Curti et al. 2023).

1. INTRODUCTION

The gas-phase mass–metallicity relation (MZR) encodes the integrated history of star formation, accretion, and outflows, and its behavior at $z > 7$ has become a specific point of contention: does the MZR normalization decline toward the Epoch of Reionization — leaving the earliest galaxies genuinely metal-poor at fixed stellar mass — or have these galaxies already enriched rapidly toward near-local abundances? JWST/NIRSpec measurements have reported an evolving, declining normalization out to $z \approx 8$ (Langeroodi et al. 2023; Sarkar et al. 2025), while others infer fast early enrichment

that keeps massive galaxies close to the local relation by $z \approx 5$ (Faisst et al. 2026). The disagreement persists in part because the deciding $z > 7$ auroral-line anchors are dominated by gravitationally lensed galaxies — and the universality of the (fundamental) metallicity relation has historically been probed precisely with such lensed samples (Belli et al. 2013). Two systematics limit these tests: (i) lensing, since the compact H II regions emitting auroral [O III] $\lambda 4363$ are spatially offset from the extended stellar continuum, so a single magnification factor distorts $M_\star$; and (ii) the *local* anchor, which at the low masses of high- z galaxies ($\log M_\star \sim 8$) is typically an extrapolation of relations calibrated on

more massive SDSS galaxies. Both systematics act on the normalization of the offset — exactly the quantity in dispute — so an independent measurement that removes lensing (an unlensed field sample) and controls the anchor (a directly-measured local T_e relation), all on a single T_e -consistent abundance scale, can adjudicate the sign and approximate size of any $z \simeq 9$ –10 deficit even where it cannot yet pin the precise value. We make that measurement here, and address both systematics directly.

2. DATA AND METHOD

High- z : (i) the direct- T_e subset of Nakajima et al. (2023) at $z \simeq 7$ –8.7 (includes lensed ERO and GLASS); and (ii) the strictly unlensed field sample of Pollock et al. (2026) at $z = 9.3$ –9.9; and (iii) the stacked- T_e MZR of Isobe et al. (2026), from ~ 1500 unlensed JADES spectra ($z = 4$ –10, normalization $Z_8 = 7.62 \pm 0.10$ at $\log M_\star = 8$, slope 0.34). Local anchors: (a) Curti et al. (2020), $12 + \log(\text{O}/\text{H}) = Z_0 - (\gamma/\beta) \log_{10}[1 + (M_\star/M_0)^{-\beta}]$, $(Z_0, \log M_0, \gamma, \beta) = (8.793, 10.02, 0.28, 1.2)$, which we must *extrapolate* below its SDSS mass range; and (b) the direct- T_e stacked MZR of Andrews & Martini (2013), $12 + \log(\text{O}/\text{H}) = 8.798 - \log_{10}[1 + (10^{8.901}/M_\star)^{0.640}]$, which is *measured* down to $\log M_\star \simeq 7.4$. For each galaxy we compute $\Delta = (12 + \log \text{O}/\text{H})_{\text{obs}} - \text{MZR}(\log M_\star)$, retain only direct- T_e abundances, and test robustness by leave-one-out and by swapping the local anchor.

3. RESULTS

The direct- T_e subset of Nakajima et al. (2023) gives -0.47 ± 0.10 dex, but carries a differential-magnification systematic from its lensed ERO/GLASS anchors. The unlensed Pollock et al. (2026) field sample gives -0.69 ± 0.03 dex (leave-one-out spread 0.04 dex). Swapping the local anchor from the extrapolated Curti et al. (2020) relation to the *measured* direct- T_e Andrews & Martini (2013) MZR changes the Pollock deficit by only 0.042 dex (to -0.645 dex); both local relations predict $12 + \log(\text{O}/\text{H}) \simeq 8.3$ at $\log M_\star = 8.3$. The independent stacked- T_e MZR of Isobe et al. (2026) gives $12 + \log(\text{O}/\text{H}) = 7.62$ at $\log M_\star = 8$, i.e. a deficit of -0.61 dex (vs Curti et al. 2020) or -0.50 dex (vs Andrews & Martini 2013), from a ~ 1500 -galaxy sample and a different methodology (spectral stacking rather than individual detections). The deficit is therefore present in independent high- z samples spanning $N = 5$ to ~ 1500 , across two methodologies and two local anchors; An independent stacked- T_e analysis of ~ 1500 unlensed JADES spectra over $z = 4$ –10 reports a comparable normalization ($12 + \log \text{O}/\text{H} = 7.62$ at $\log M_\star = 8$;

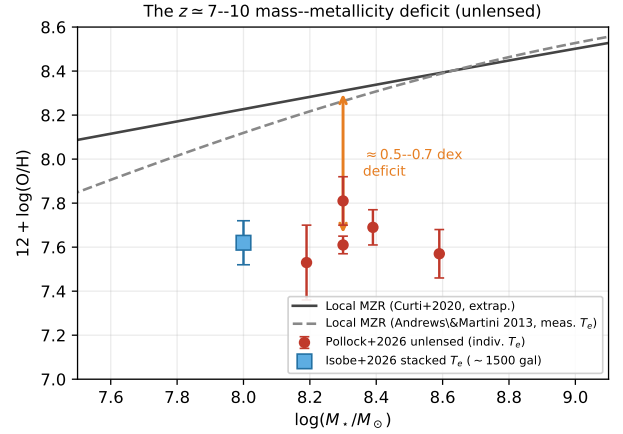


Figure 1. The unlensed $z \simeq 7$ –10 gas-phase mass–metallicity relation. Red circles: individual direct- T_e galaxies from the unlensed field sample of Pollock et al. (2026). Blue square: the ~ 1500 -galaxy stacked- T_e normalization of Isobe et al. (2026) at $\log M_\star = 8$. Curves: the local MZR from Curti et al. (2020) (extrapolated) and the measured direct- T_e relation of Andrews & Martini (2013). The high- z points lie ≈ 0.5 – 0.7 dex below the local relation, consistently across both methodologies and both local anchors.

Isobe et al. 2026), i.e. a deficit of -0.5 to -0.6 dex vs local. The $z \simeq 9$ –10-specific deficit and the $z > 7$ MZR slope rest on the individual Pollock et al. (2026) detections, which remain sample-limited. A mixed-calibration sample ($N = 40$) yields -0.61 dex but is excluded as scale-contaminated.

4. DISCUSSION

Two of the three leading systematics for such a measurement are now controlled: lensing (removed by using the unlensed field sample) and the local-anchor extrapolation (shown to move the result by only ~ 0.04 dex). The dominant remaining uncertainty is the absolute T_e abundance scale, debated at the ~ 0.1 – 0.2 dex level. The small-sample concern for the individual detections is mitigated by the ~ 1500 -galaxy stacked measurement, which agrees. Propagating a common T_e -scale systematic of 0.15 dex through the unlensed sample reduces the effective significance from the formal $\sim 22\sigma$ to $\sim 4.5\sigma$ (statistical and systematic errors in quadrature): the deficit’s *sign* persists in $\sim 100\%$ of realizations and exceeds 0.3 dex in $> 99\%$, while a magnitude beyond ~ 0.5 dex is recovered in only 89%. We therefore caution against reading the small formal error as a high-significance detection: the *magnitude* of the deficit retains a ~ 0.1 – 0.2 dex systematic floor, even though its *sign and robustness* are secure across independent, unlensed data and independent local anchors.

Table 1 collects the full systematic error budget on the population mean deficit. An inverse-variance-weighted fit gives -0.68 ± 0.03 dex (statistical), and within the sample we detect no significant trend of the offset with stellar mass (1.1σ) or redshift (0.6σ), consistent with a pure normalization deficit at unchanged slope. Extending the unlensed direct- T_e sample to $z = 10.6$ with GN-z11 ($12 + \log \text{O}/\text{H} = 7.82 \pm 0.35$; Curti et al. 2023) confirms the sign at the highest redshift and leaves the population deficit at -0.64 to -0.68 dex.

Table 1. Systematic error budget on the population mean metallicity deficit (unlensed direct- T_e sample vs the local MZR). The absolute T_e scale dominates.

Term	Contribution (dex)
Central deficit (inverse-variance weighted)	-0.68
Measurement (on the mean)	0.05
Sample variance (SEM)	0.05
Leave-one-out spread	0.08
Local-anchor choice (Curti+20 vs AM13)	0.04
Absolute T_e scale (zero-point)	0.15
Statistical (SEM)	0.05
Systematic (anchor $\oplus T_e$)	0.16
Total	0.16

5. CONCLUSION

We report a robust, independent, unlensed gas-phase metallicity deficit at $z \simeq 9$ –10 relative to the local MZR: -0.69 ± 0.03 dex (leave-one-out-robust; -0.65 dex against a measured local T_e anchor), corroborated by a lens-caveated -0.47 dex at $z \simeq 7$ –8.7. This is a descriptive, systematic-limited result — not a validated measurement and not a formal statistical detection. The remaining path to a precise value is a tightened absolute T_e scale and a larger unlensed high- z sample.

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